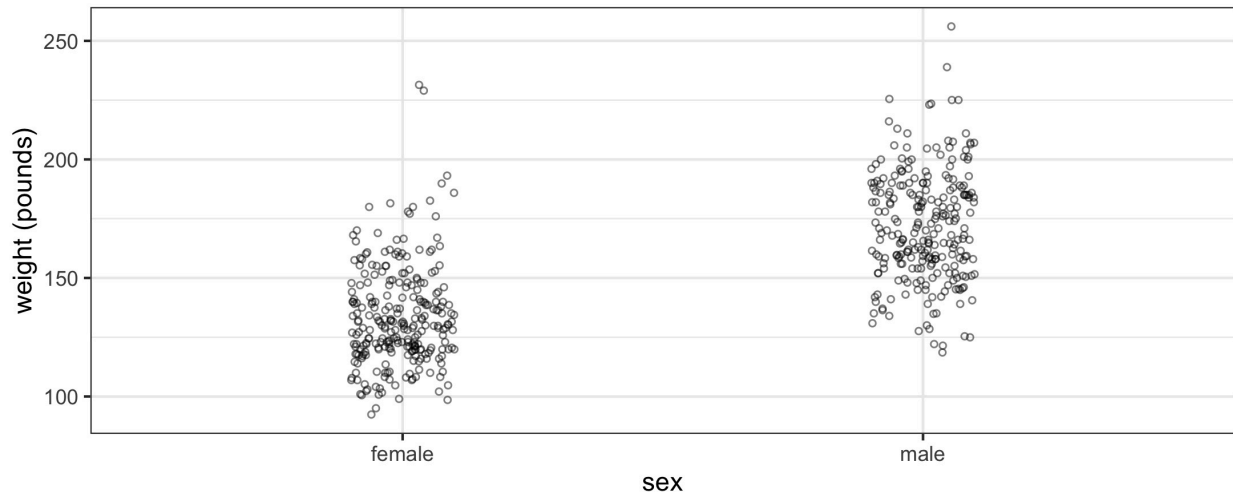


Quiz 5

Names of group members: Moyed Imtiaz, Daniel Coblentz, Tommy Zanchettin

In this quiz, you will look again at the data set from last week with body measurements from 507 physically active US adults. In our earlier work, we focused on predicting weights using heights alone.

I made a graph in R of the weights of people in the sample grouped by sex.



I also used “lm” to model weights using sex alone.

Call:

```
lm(formula = weight ~ sex, data = body_dims)
```

Residuals:

Min	1Q	Median	3Q	Max
-53.338	-13.958	-3.081	13.982	98.119

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	133.321	1.373	97.11	<2e-16 ***
sexmale	38.597	1.967	19.62	<2e-16 ***

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 22.14 on 505 degrees of freedom

Multiple R-squared: 0.4326, Adjusted R-squared: 0.4315

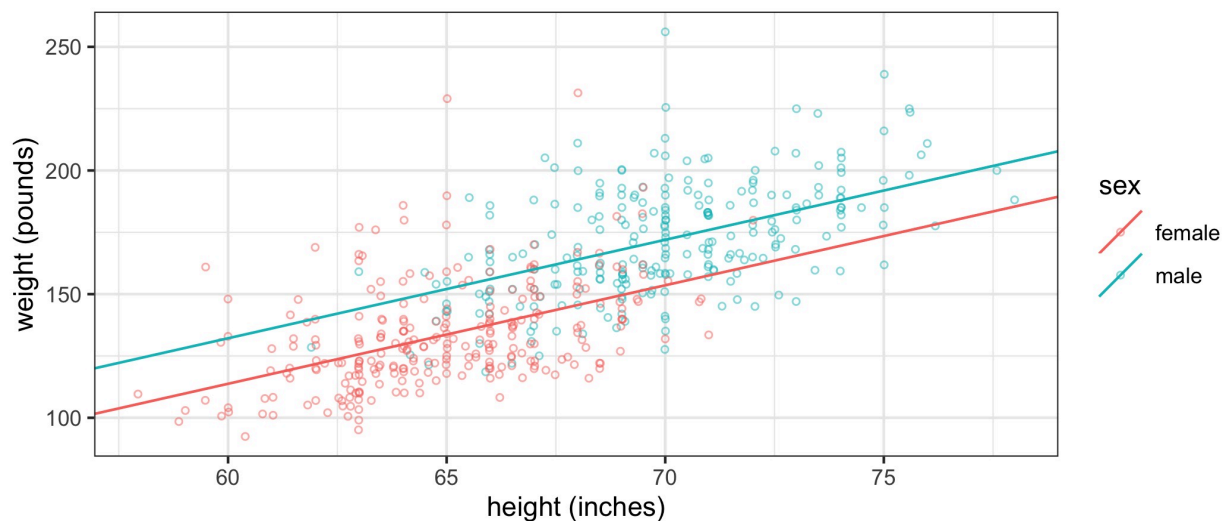
F-statistic: 385 on 1 and 505 DF, p-value: < 2.2e-16

1. Use the model summary to complete the following table. (You can check your numbers roughly by looking at the scatter plot above.)

Mean weight for females in sample (pounds)	133.321 lbs
Mean additional weight for males in sample (pounds)	38.597 lbs
Mean weight for males in sample (pounds) (Note that you can get that from adding the above two numbers)	171.918 lbs

We know that there are multiple factors contributing to the fact that males tend to weigh more than females. On the one hand, males tend to be bulkier than females. On the other hand, males also tend to be taller than females.

To sort out these two effects, we can use a regression model with both sex and height as predictors. I have included below a graph of the model and the summary of the model from R.



```
Call:
lm(formula = weight ~ sex + height, data = body_dims)

Residuals:
    Min       1Q   Median       3Q      Max
-44.404 -13.152  -2.983   10.361   95.342

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -125.2889    20.7338  -6.043 2.95e-09 ***
sexmale      18.4052     2.3605   7.797 3.66e-14 ***
height        3.9841     0.3189  12.494 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 19.36 on 504 degrees of freedom
Multiple R-squared:  0.5668,    Adjusted R-squared:  0.5651
F-statistic: 329.7 on 2 and 504 DF,  p-value: < 2.2e-16
```

2. Write the equation for the model. (There should be a “hat” somewhere in what you write.)

$\text{Weight_hat} = -125.2889 + 3.9841(\text{height_in}) + 18.4052(\text{sexmale})$

***note** height_in is height in inches and sexmale is binary for male(1) or female(0)

3. What does the model predict for the weight of a 68-inch-tall male?

$\text{Weight_hat} = -125.2889 + 3.9841(68) + 18.4052(1)$

weight_hat = 164.0351 lbs

4. Write a sentence interpreting the coefficient of “sexmale” in context.

When the gender changes from female to male, the weight increases by 18.4052 lbs.

5. Write a sentence interpreting the coefficient of “height” in context.

For every inch of someone’s height, the weight increases by 3.9841 lbs.

6. The coefficient of “sexmale” is smaller in the model that includes height as well as sex as predictors. Why does that make sense?

In the first model, there is only one factor that affects the weight of a person, meaning that coefficient would be greater to compensate for all possible factors contributing to weight.

In the second model, there are two factors that affect the weight of a person, meaning that the sex coefficient would be less, as the burden of determining weight is distributed on two variables. Females who are in a similar height category to males are going to have a relatively similar weight therefore decreasing the strength of sexmale variable on the model.